

InvoraLite — Product Quality Review

Review date: 4 July 2026 | **Exported:** 04|07|2026 | **Version:** 1.0.0 | **Stack:** Tauri 2 + React 19 + SQLite

Executive Summary

Overall Rating: 8.4 / 10

InvoraLite is a mature v1.0 desktop inventory application for a single store, with unusually broad coverage for a “lite” product: sales, purchases, credit, supplier advances, GST invoices, analytics, double-entry-style accounting, tax submission reports, and a complete print ecosystem for invoices and receipts.

Since the prior review, the product gained **database encryption at rest** (AES-256-GCM vault sealed on exit, key protected via Windows DPAPI), **encrypted backup exports** (password-protected portable v2 archives with Argon2id; machine-key auto backups), and **full frontend role-based UI gating** across all pages and admin panels. Earlier releases added sale voids, stock adjustments, low-stock alerts, audit log viewer, CSV import/export, supplier ID on purchases, receipt previews, customer payment receipts (A5 landscape), and tax submission printing.

The app is **ready for trusted single-store deployment** on Windows. Remaining gaps are mainly operational and scale-related: license secrets in source, JSON-blob data model, no E2E/CI pipeline, and no code signing or auto-update.

Rating Breakdown

Area	Score	Notes
Feature breadth	8.5/10	Sales, purchases, stock, returns, adjustments, people, credit, invoices, analytics, accounting, tax reports, office assets, encrypted backup, licensing
UX / polish	7.8/10	Cohesive Invora theme, modals, print previews, role guidance banners, access-restricted panels; some large page files remain
Business logic	8/10	Supplier AP, customer credit FIFO, purchase→stock, accounting journal, tax report generation
Security	8/10	Argon2, RBAC (backend + frontend), lockout, audit, DB-at-rest encryption, encrypted backups; license secrets still in source
Data / reliability	6.5/10	Health checks, encrypted vault, backup/restore with manifest; JSON-in-SQLite limits integrity at scale
Testing & quality	5.8/10	27 Rust + 23 Vitest tests (50 total); no E2E, no CI, no component tests
Deployment	7.5/10	Windows NSIS installer with dated export naming; user manual + product review PDFs; no code signing or auto-update

Technology Stack

Layer	Technology
Desktop shell	Tauri 2, WebView2
Frontend	React 19, TypeScript 5.6 (strict), Vite 6, Tailwind 4
Charts	Recharts 2.15
Backend	Rust 2021 — src-tauri/src/
Database	SQLite via rusqlite 0.32 (bundled, WAL, backup API)
Encryption	AES-256-GCM (vault), Argon2id (backup passwords), Windows DPAPI (key file)
IPC	Tauri commands (storage, audit, auth, license, backup, password)

Architecture

```
React (pages, components, lib/data.ts, usePermissions hook)
  → storage.ts (passes username + role as StorageContext)
    → Tauri commands
      → DatabaseManager (src-tauri/src/db/manager.rs)
        → validation · sql_guard · rbac · storage_repository · audit_repo
        → encryption_key.rs (seal/unseal invora.db.vault on shutdown)
        → backup_archive.rs (v1 plaintext + v2 encrypted restore)
```

Data model: Business entities live as JSON documents in app_storage. System tables (audit_log, auth_lockout, schema_meta) are relational.

Database path: %LOCALAPPDATA%/InvoraLite/ — live DB unsealed at runtime; sealed to invora.db.vault when the app exits.

Bootstrap flow: Splash → License gate → Setup → Login → Main layout (with role banner).

Feature Completeness

Implemented (substantial)

Module	Highlights
License	60-day trial, device-bound keys, ZIP activation
Auth / setup	Business profile, GST, staff accounts, 30-min session timeout
Dashboard	Stats, charts, low-stock alerts, quick new-sale CTA (role-gated)
Products	CRUD, categories, IMEI fields, low-stock threshold, CSV import/export, stock adjustments tab
Stock adjustments	Stocktake, damage, theft with audit trail
Sale returns	Void sales, restore stock, reverse credit (Products tab + Invoice page)
Purchases	Multi-line, GST, supplier ID, receive-to-stock (Admin + Store Keeper only)
Sales	Cash, e-payment, credit, partial payment; thermal receipt preview & print
People	Customers & suppliers (tab visibility by role), credit, payments, A5 payment receipt
Invoices	Fixed sales panel, divider layout, print, credit settlement, void (role-gated)
Analytics	Weekly / monthly / yearly earnings
Accounting	Chart of accounts, journal, P&L, monthly close, depreciation, tax submission report
Office	Expenses, assets, expense print reports (Admin only)
Settings	Staff roles, business profile (read-only for non-Admin), audit log, encrypted backup export/restore
Backup	Password-encrypted manual export (portable) + machine-key auto startup backups (keep 10); v1 plaintext restore supported
RBAC UI	Sidebar, pages, modals, and admin panels hide/disable actions by role; AccessRestricted + ReadOnlyBanner
Documentation	User manual PDF, product review PDF, dated installer exports

Remaining gaps

- Multi-store / cloud sync — not supported (single-computer design)
- Code signing and auto-update — not configured
- License secrets in source — hardcoded signing key and ZIP password need rotation before wide distribution
- E2E and CI pipeline — not present
- macOS / Linux bundles — Windows NSIS only
- Relational schema — core entities still stored as JSON blobs in app_storage

Role enforcement Complete (v1.0.0)

Role	Products	Purchases	Sales	Customers	Suppliers	Delete	Admin settings
Admin	✓	✓	✓	✓	✓	✓	✓
Store Keeper	✓	✓	✓	✓	✓	✗	Business view only
Cashier	View	✗	✓	✓	✗	✗	Business view only
Viewer	View	✗	View	View	View	✗	Business view only

Backend (Rust): All writes blocked at storage layer for unauthorized roles. **Frontend:** Navigation, tabs, forms, and destructive actions gated via usePermissions; Purchase page and admin panels show AccessRestricted when denied.

Security Posture**Strengths**

Control	Implementation
Centralized DB access	DatabaseManager — UI never touches SQLite directly
Database encryption at rest	AES-256-GCM vault (invora.db.vault); key in .dbkey via Windows DPAPI; sealed on app exit
Encrypted backups	v2 archives: Argon2id + AES for portable exports; machine-key for auto backups; v1 plaintext still restorable
Parameterized SQL	All storage queries use bound parameters
RBAC	Role-based write/delete on storage keys; mirrored in React UI
Password hashing	Argon2id; legacy plaintext migrated on login
Account lockout	5 failures → 15 minutes
Audit trail	Login, stock changes, sale voids — viewable in Manage → Audit Log
Backups	Consistent snapshots, manifest verification, restore with rollback
Session timeout	30 minutes idle logout
CSP	Content Security Policy configured in tauri.conf.json

Remaining risks

Risk	Severity	Detail
License secrets in source	High	LICENSE_SECRET and ZIP password need rotation before wide distribution
Runtime plaintext DB	Low–Medium	DB is decrypted in memory and as invora.db while the app is running
Machine-key backups	Low	Auto backups only restorable on the same Windows user account
No code signing	Medium	Installer may trigger SmartScreen warnings

Testing Coverage

Rust — 27 tests (cargo test): SQL injection, RBAC, rollback, lockout, integrity, Argon2 migration, encryption roundtrips, backup archive v1/v2.

Frontend — 23 tests (npm test): payment balances, password policy, credit helpers, FIFO allocation, permissions.

Gaps: No component tests, E2E, license/accounting UI tests, or CI pipeline.

Print & Reporting

Document	Format
Sale receipt	80mm thermal — preview then print
Tax invoice	A4 — from Invoice page
Customer payment receipt	A5 landscape — preview then print; shows invoice FIFO + balance
Tax submission report	A4 portrait — P&L, GST, sales & purchase registers
Office expense report	Standard — from Manage → Office
User manual	PDF — docs/InvoraLite_User_Manual.pdf
Product review	PDF — docs/InvoraLite_Product_Review.pdf

Build & Deployment

Item	Detail
Dev	npm run tauri:dev
Build	npm run tauri:build → NSIS installer
Output	exports/InvoraLite v.1.0.0 DD-MM-YYYY x64-setup.exe
Docs export	npm run review:pdf / npm run docs:pdf
Targets	Windows NSIS (x64)

Not configured: code signing, auto-update, macOS/Linux bundles, release CI.

Recommended Improvements

Before wider distribution

1. Replace license secrets; remove hardcoded ZIP password
2. Code-sign the Windows installer
3. CI running npm test + cargo test on every release

Medium term

4. Relational schema migration for core entities
5. E2E smoke tests for sale, purchase, backup, and role-restriction flows
6. Auto-update channel for patch releases
7. macOS bundle for cross-platform clients

Changes Since Last Review

Item	Prior status	Current status
Database encryption	Not active	Done — AES-256-GCM vault + DPAPI key
Encrypted backups	Plaintext ZIP	Done — v2 password + machine-key auto backups
Frontend RBAC	Partial	Done — all pages, tabs, and admin panels gated
Overall rating	8.0 / 10	8.4 / 10
Security score	6.5 / 10	8.0 / 10

Use-Case Fit

Scenario	Fit
Owner's own shop, trusted staff	9/10 — strong daily tool with accounting, tax reports, and role separation
Few clients, supervised rollout	8/10 — viable with license rotation + encrypted backup discipline
Many businesses / sensitive data	6.5–7/10 — encryption and RBAC help; still needs license hardening and CI
Multi-store / enterprise	4/10 — JSON model and single-PC design not ready

Top Strengths

1. Real shop workflows — credit sales, partial payments, supplier advances, Bhutan e-payments, GST, IMEI tracking
2. Security foundation — encrypted DB at rest, encrypted backups, centralized Rust data layer, full RBAC
3. Accounting depth — journal, P&L, monthly close, printable tax submission report

4. Offline-first desktop — local SQLite, backup/restore, dated Windows installer
5. Complete print ecosystem — invoices, thermal receipts, payment receipts, tax reports
6. Polished product identity — consistent UI, themes, role guidance, documentation PDFs

Top 3 Next Steps

1. **Production license hardening** — rotate secrets, externalize signing key
2. **Code signing + CI** — signed installer and automated test gate before each release
3. **E2E smoke tests** — sale, purchase, backup restore, and role-restriction flows

InvoraLite Product Quality Review v1.0.0 — Exported 04/07/2026 — EDP IT Department. See also docs/DATABASE_ARCHITECTURE.md and docs/InvoraLite_User_Manual.pdf.